



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	MECHANICS -II				
Type of Course	Core in Major				
Semester	IV				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	PHY3CJ201: Mechanics -I				
Course Summary	This course explores Newton's Laws of Motion and how they can be applied to solve different mechanical systems.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Explain the principles of central force motion and derive key results related to Kepler's laws and planetary motion.	U	C	Instructor-created exams / Quiz
CO2	Apply Newton's laws and the work-energy theorem to solve problems involving rotational and oscillatory motion.	Ap	P	Instructor-created exams / Home Assignments

CO3	Analyze the dynamics of damped and forced harmonic oscillators.	An	C,P	Instructor-created exams / Home Assignments
CO4	Analyze travelling and standing waves and the transport of energy in wave motion.	An	C	Instructor-created exams / Home Assignments
CO5	Evaluate effects of accelerated and rotating frames, including fictitious forces.	E	C,P	Seminar Presentation / Group Tutorial Work
CO6	Design and perform experiments and computational simulations related to mechanics, using smartphone-based sensors and Python tools.	Ap	P	Practical Assignment / Observation of Practical Skills / Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F), Conceptual Knowledge (C), Procedural Knowledge (P), Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	CENTRAL FORCE MOTION: THE KEPLER PROBLEM		14	20
	1	Kepler's Laws	1	
	2	Central Forces	2	
	3	The Equation of Motion	2	
	4	Energy and the Effective Potential Energy	2	
	5	Solving the Radial Equation of Motion	1	
	6	The Equation of the Orbit	2	
	7	The Equation of an Ellipse	2	
	8	Kepler's Laws Revisited	2	

	Sections 10.1 – 10.9 of chapter 10 of Book 1			
II	HARMONIC MOTION		13	20
	9	Springs and Pendulums	1	
	10	Solving the Differential Equation – Undamped Harmonic Oscillator	2	
	11	Damped Harmonic Oscillator – Underdamped, Overdamped and Critically Damped Oscillators	4	
	12	The Forced Harmonic Oscillator – Forced Undamped and Forced Damped Oscillators	4	
	13	The Q Factor	1	
	14	Resonance in Electrical Circuits	1	
	Sections 11.1 – 11.4 of chapter 11 of Book 1			
III	WAVES		8	14
	15	A Wave in a Stretched String	1	
	16	Direct Solution of the Wave Equation	1	
	17	Fourier Series	1	
	18	Standing Waves and Traveling Waves	2	
	19	Standing Waves as a Special Case of Traveling Waves	1	
	20	Energy and Energy Flow	2	
	Sections 13.1 – 13.6 of chapter 13 of Book 1			
IV	ACCELERATED REFERENCE FRAMES		10	16
	21	A Linearly Accelerating Reference Frame	1	
	22	A Rotating Coordinate Frame	1	

	23	Fictitious Forces	2	
	24	Centrifugal Force and the Plumb Bob	1	
	25	The Coriolis Force – A Falling Body and A Projectile	3	
	26	The Foucault Pendulum	2	
	Sections 15.1 – 15.6 of chapter 15 of Book 1			
V	PRACTICALS		30	
	Conduct any 6 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. • The necessary theory of the experiments can be given as an Assignment/ Seminar. • Calculate the percentage error and standard deviation in each experiment. • Plot the graphs using Python. • Smartphones are exclusively intended for educational lab use. Necessary care should be taken to safeguard them during the experiments. • Smartphone experiments primarily serve demonstration purposes, with result accuracy contingent upon the precision of phone sensors and experimental setups.			
	1	Flywheel- Determination of the Moment of Inertia. • This experiment aims to help students grasp the concept of energy conservation and the dynamics of rotation. • Do at least 9 trials for different masses and number of turns wound on the axil.		
	2	Torsion Pendulum- Determination of the Moment of Inertia and Rigidity Modulus. • Using identical masses on the disc, determine the moment of inertia of the disc. • Verify the moment of inertia by direct method, $I = \frac{1}{2}MR^2$ • Using I, calculate rigidity modulus of the material of the wire, $n = \frac{8\pi I}{r^4} \frac{L}{T^2}$		
	3	Compound Pendulum- Acceleration Due to Gravity and Moment of Inertia and Verification of Parallel Axis Theorem.		

		- Plot a graph of distance of knife edge from one end Vs period of oscillations. Using the measurement from the graph, calculate g. - Calculate the radius of gyration and hence the moment of inertia about CM. Compare the result obtained by the direct calculation $I_{CM} = \frac{ML^2}{12}$ - Measure the period of oscillation about an arbitrary pivot point which is at a distance d from the CM. Calculate $I_{pivot} = mgd \frac{T^2}{4\pi^2}$. - Verify the result using parallel axes theorem, $I_{pivot} = I_{CM} + md^2$		
4	Kater's Pendulum- Determination of Earth's Gravity.	To determine g for both the cases (a) $T_1 \approx T_2$ and (b) $T_1 \neq T_2$ and discuss the relative merits of both cases by estimation of error in the two cases.		
5	Melde's String - Determination of the Frequency of the Turing Fork	- Determine the frequency of electrically maintained tuning fork by means of Melde's apparatus in longitudinal and transverse mode of vibration. - Verify $\lambda^2 - T$ law.		
6	Sonometer - Determine the Frequency of AC.	- Estimate the linear mass density of the wire. - Draw $L^2 - m$ graph and from the slope calculate the frequency.		
7	Fourier Analysis of the Modes of Vibration in a Stretched String.	- Record the sound produced by guitar string (or similar arrangement) using a microphone and analyze the spectrum by taking Fast Fourier Transform (FFT). - Audio Spectrum in the Pyphox, Audacity, ExpEYES or any other tools can be used to record the sound and get the FFT. - Vary the length and tension of the string and analyze the harmonics. https://phyphox.org/experiment/audio-spectrum/ https://www.youtube.com/watch?v=bl7jf2myEvM https://expeyes.in/experiments/sound/beats.html		
8	Determination of the Velocity of Sound in Air.	Sound wave of known frequency is generated using a wave generator(WG) and piezo buzzer and are recorded using a microphone(MIC).		

		- Phase differences between the WG and MIC waveforms were analyzed in a CRO and the distance between them were adjusted to make both of them in phase and hence calculate velocity of sound. - Phase difference can be analyzed from the Lissajous figure obtained by X-Y plotting of WG and MIC waves. - ExpEYES may be used. https://expeyes.in/experiments/sound/velocity.html https://expeyes.in/experiments/electrical/xyplot.html		
	9	Transformation of Energy from One Form to Another. - Roll a hollow cylinder from a height, in an inclined plane, without pushing. - Measure radius of the cylinder and record the velocity of the cylinder using the gyroscope of the phone inserted into the cylinder. - Calculate the total energy before the cylinder starts to roll (Potential Energy, mgh) - Calculate the total energy (Translational KE + Rotational KE) when the cylinder reaches the bottom of the plane. - Repeat the experiment for different heights. - Experiment 23 for Book 2 https://phyphox.org/experiment/roll/#more-509		
	10	Pendulum- Limits on Angular Displacement and Study of Damped Oscillations. - Estimate limits on angular displacement for SHM by measuring the time period at different angular displacements and compare it with the expected value of time period for SHM. Example 12.1 of Book 1. - Study damped oscillations. Plot amplitude as a function of time and determine the damping coefficient and Q factor. - Digitized data can be used for the study. https://www.youtube.com/watch?v=jcpvm95bhXw https://expeyes.in/experiments/school-level/sr04.html https://phyphox.org/experiment/pendulum/		
	11	Realize the computational Projects in chapters 10, 11, 12, 13, 15 of Book 1 or any other related projects using Python		

Books and References:

- Intermediate Dynamics (Edn.2) by Patrick Hamill (Book 1)
- Smartphones as Mobile Minilabs in Physics(Edn. 1) by Jochen Kuhn & Patrik Vogt, Springer, (Book 2)
- An Introduction to Mechanics by Daniel Kleppner and Robert J. Kolenkow

4. Mechanics by Keith R. Symon
5. Mechanics: Berkeley Physics Course, Volume 1 by Charles Kittel, Walter D. Knight and Malvin A. Ruderman
6. Mechanics: From Newton's Laws to Deterministic Chaos by Florian Scheck
7. NPTEL video lectures: <https://nptel.ac.in/courses/115106090>

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	0	2	0	0	0	2	0	0	0	0	0	0
CO 2	2	2	2	0	0	0	2	2	0	0	0	0	0
CO 3	0	2	2	0	0	0	0	2	0	0	0	0	0
CO 4	0	2	2	2	0	0	0	2	2	0	0	0	0
CO 5	0	0	2	0	0	0	0	0	2	0	0	0	0
CO 6	0	2	2	2	0	2	0	2	2	0	0	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	